Thirsty Engines: Why Your 1974 Car Hates Your Wallet

Jozef Rivest

University of Michigan

Motivation

Humanity's oldest question: *why am I at the gas station again?* Your uncle with the 1973 Cadillac has opinions. This poster has data (Arel-Bundock 2025).

Puzzle

The Toyota Corolla gets **33.9** miles per gallon. The Cadillac Fleetwood gets **10.4**. What is the Cadillac *doing* with all that gas?

A bit of theory

Two mechanisms explain engine thirst:

1. **Many Mouths:** each cylinder is a tiny mouth that drinks gasoline
2. **Enthusiasm:** horsepower is *how hard* those mouths drink

H1: More cylinders mean fewer miles per gallon.

H2: Holding cylinders constant, more horsepower means fewer miles per gallon.

Background

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Related work

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Scope conditions

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Design

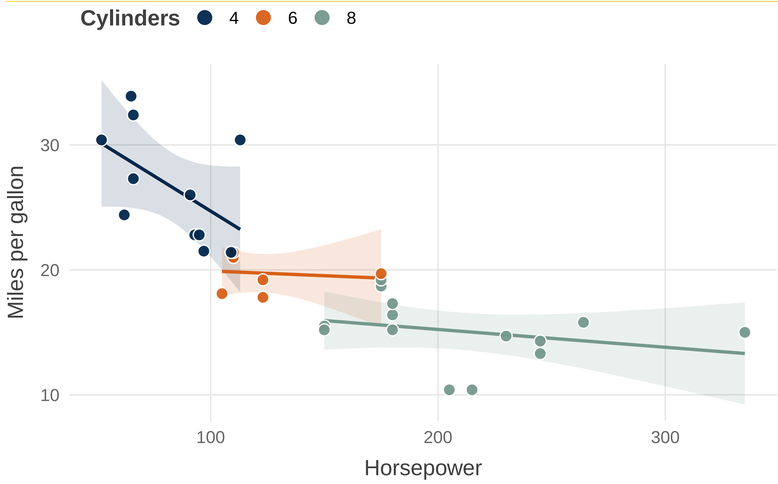
I estimate the following model with OLS, because I know how to do it:

$$\text{MPG}_i = \beta_0 + \beta_1 \text{Cyl}_i + \beta_2 \text{HP}_i + \varepsilon_i$$

Data

- `mtcars`: 32 cars road-tested by *Motor Trend* in 1974
- Outcome: miles per gallon (higher = smugger)
- Predictors: cylinders (mouths) and horsepower (enthusiasm)

Results 1



Robustness

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Heterogeneity

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Results 2

H1 confirmed: an 8-cylinder car loses about 8.5 mpg relative to a 4-cylinder car.

	Model 1
Constant	28.650***
	(1.588)
6 cylinders	-5.968**
	(1.639)
8 cylinders	-8.521**
	(2.326)
Horsepower	-0.024
	(0.015)
Num.Obs.	32
R2	0.754

Conclusion

- More cylinders, fewer miles per gallon
- **H2 runs out of gas:** horsepower is guilty in spirit, but $p \approx 0.13$
- Future work: data from after 1974, and asking who *needs* a V8 for groceries

Vroom?

Next steps

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Acknowledgements

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References

Arel-Bundock, Vincent. 2025. *Model to Meaning: How to Interpret Statistical Models with r and Python*. 1st ed. Chapman; Hall/CRC. `https://doi.org/10.1201/9781003560333`.